

There are six problems in total. You must solve the first five, and problem 6 is optional.

Problem 1 (CO5): Pumping Lemma (5 points)

Consider the following language. [Recall: For any string w , $|w|$ denotes the length of w .]

$$L = \{0^i 1w : w \in \{0,1\}^* \text{ and } i \geq |w|\}$$

Prove that L is not regular using the pumping lemma.

Problem 2 (CO3): Designing Context-Free Grammars (15 points)

Consider the following languages.

$$L_1 = \{w \in \{a,b\}^* : w \text{ is a palindrome}\}$$

$$L_2 = \{w \in \{a,b\}^* : w \text{ contains } bb \text{ as a substring}\}$$

$$L_3 = \{w \in \{a,b\}^* : w \text{ does not contain any palindromic substring of length two}\}$$

$$L_4 = L_1 \cap L_3$$

$$L_5 = \{w_1 \# w_2 : w_1, w_2 \in L_2 \text{ and } |w_1| = |w_2|\}$$

Now solve the following problems.

- (a) **Give** a context-free grammar that generates L_1 . (3 points)
- (b) **Give** a context-free grammar that generates L_2 . (3 points)
- (c) **Give** a context-free grammar that generates L_3 . [Note: A palindromic substring is defined as a substring that reads the same backward as forward. For example, in the string 'abab' – 'ε', 'a', 'b', 'aba', 'bab' are palindromic substrings.] (3 points)
- (d) **Write** all five-letter strings in L_4 . (1 point)
- (e) **Give** a context-free grammar that generates L_4 . (2 points)
- (f) **Give** a context-free grammar that generates L_5 . (3 points)

Problem 3 (CO3): Derivations, Parse Trees and Ambiguity (10 points)

Consider the following context-free grammar.

$$S \rightarrow DQ$$

$$D \rightarrow Daa \mid Dab \mid Dba \mid Dbb \mid a \mid b$$

$$Q \rightarrow aQb \mid bQa \mid \epsilon$$

- (a) **Give** a leftmost derivation for the string 'abbababab'. (3 points)
- (b) **Draw** the parse tree corresponding to the derivation you gave in (a). (2 points)
- (c) **Show** that the given grammar is ambiguous by drawing two more parse trees (apart from the one that you have drawn in (b)) for the given string in (a). (4 points)
- (d) **Write** one five-letter string that has exactly one parse tree in the given grammar. (1 point)

Problem 4 (CO3): Designing Pushdown Automata (15 points)

Consider the following languages.

$$L_1 = \{0^i 1^j 2^k : j = i + k \text{ or } i = j + k; \text{ where } i, j, k \geq 0\}$$

$$L_2 = \{w_1 \# w_2 : w_1, w_2 \in \{0, 1\}^* \text{ and } |w_1| \neq |w_2|\}$$

$$L_3 = \{w_1 \# w_2 : w_1, w_2 \in \{0, 1\}^*, w_1 \text{ contains odd numbers of 0s and } |w_1| = |w_2|\}$$

Now solve the following problems.

- (a) **Give** the state diagram of a pushdown automaton that recognizes L_1 . (5 points)
- (b) **Give** the state diagram of a pushdown automaton that recognizes L_2 . (5 points)
- (c) **Give** the state diagram of a pushdown automaton that recognizes L_3 . (5 points)

Problem 5 (CO4): Turing Machines (5 points)

- (a) For any Turing-recognizable language L , is it possible to build a recognizer for $\bar{L}$? [Recall: $\bar{L}$ denotes the complement of the language L] (1 point)

Consider the following language.

$$L_1 = \{0^n \# 1^n \# 2^m : n, m \geq 0 \text{ and } n < m\}$$

- (b) **Give** the state diagram of a turing machine that decides L_1 . (4 points)

Problem 6 (Bonus): Designing Turing Machines (3 points)

Consider the following language. [Note: $\text{Count}(w, x)$ is defined as the count of x in the string w . For example, $\text{Count}(\text{abba}, b) = 2$.]

$$L = \{w_1 \# w_2 \# w_3 : w_1, w_2, w_3 \in \{a, b, c\}^* \text{ and } \text{Count}(w_3, c) = \text{Count}(w_1, a) \times \text{Count}(w_2, b)\}$$

Give the state diagram of a turing machine that decides L .

After you are finished with the exam, please indicate where you stand on the smiley face spectrum.



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Problem 1 (CO5): Pumping Lemma (5 points)

Consider the following language. [Recall: For any string w , $|w|$ denotes the length of w .]

$$L = \{w10^i : w \in \{0,1\}^*, i \geq 0 \text{ and } |w| \geq i\}$$

Prove that L is not regular using the pumping lemma.

Problem 2 (CO3): Designing Context-Free Grammars (15 points)

Consider the following languages.

$$L_1 = \{w \in \{0,1\}^* : w \text{ is a palindrome}\}$$

$$L_2 = \{w \in \{0,1\}^* : w \text{ contains } 00 \text{ as a substring}\}$$

$$L_3 = \{w \in \{0,1\}^* : w \text{ does not contain any palindromic substring of length two}\}$$

$$L_4 = L_1 \cap L_3$$

$$L_5 = \{w_1\#w_2 : w_1, w_2 \in L_2 \text{ and } |w_1| = |w_2|\}$$

Now solve the following problems.

- (a) **Give** a context-free grammar that generates L_1 . (3 points)
- (b) **Give** a context-free grammar that generates L_2 . (3 points)
- (c) **Give** a context-free grammar that generates L_3 . [Note: A palindromic substring is defined as a substring that reads the same backward as forward. For example, in the string '0101' - ' ϵ ', '0', '1', '010', '101' are palindromic substrings.] (3 points)
- (d) **Write** all five-letter strings in L_4 . (1 point)
- (e) **Give** a context-free grammar that generates L_4 . (2 points)
- (f) **Give** a context-free grammar that generates L_5 . (3 points)

Problem 3 (CO3): Derivations, Parse Trees and Ambiguity (10 points)

Consider the following context-free grammar.

$$S \rightarrow DQ$$

$$D \rightarrow Daa \mid Dab \mid Dba \mid Dbb \mid a \mid b$$

$$Q \rightarrow aQb \mid bQa \mid \epsilon$$

- (a) **Give** a leftmost derivation for the string 'aabbababa'. (3 points)
- (b) **Draw** the parse tree corresponding to the derivation you gave in (a). (2 points)
- (c) **Show** that the given grammar is ambiguous by drawing two more parse trees (apart from the one that you have drawn in (b)) for the given string in (a). (4 points)
- (d) **Write** one five-letter string that has exactly one parse tree in the given grammar. (1 point)

Problem 4 (CO3): Designing Pushdown Automata (15 points)

Consider the following languages.

$$L_1 = \{0^i 1^j 2^k : j = i + k \text{ or } k = i + j; \text{ where } i, j, k \geq 0\}$$

$$L_2 = \{w_1 \# w_2 : w_1, w_2 \in \{0, 1\}^* \text{ and } |w_1| \neq |w_2|\}$$

$$L_3 = \{w_1 \# w_2 : w_1, w_2 \in \{0, 1\}^*, w_1 \text{ contains even numbers of 0s and } |w_1| = |w_2|\}$$

Now solve the following problems.

- (a) **Give** the state diagram of a pushdown automaton that recognizes L_1 . (5 points)
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- (a) For any Turing-recognizable language L , is it possible to build a recognizer for $\bar{L}$? [Recall: $\bar{L}$ denotes the complement of the language L] (1 point)

Consider the following language.

$$L_1 = \{1^n \# 0^n \# 2^m : n, m \geq 0 \text{ and } n < m\}$$

- (b) **Give** the state diagram of a turing machine that decides L_1 . (4 points)

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